water in the Earth's oceans. By comparing the Moon's steady motion to the Earth's slowing spin, astronomers had incorrectly inferred a lunar speedup. See Smart (1953).

96 [review article . . . *Radioaktivität und Elektronik*] An English translation of Einstein's beautiful review article is published as Document 47 of ECP-2.

100 [Einstein discovered gravitational time dilation . . . presented in Box 2.4,] Einstein's argument as presented in Box 2.4 was originally published in Einstein (1911).

100 [When starting to write his 1907 review article, . . . *Radioaktivität und Elektronik*] Document 47 of ECP-2.

103 [Einstein's life as a professor . . . he was brilliant.] See Frank (1947), pages 89–91.

117 [These conclusions . . . on 25 November.] Einstein (1915).

118–119 Box 2.6: Remark for readers who are familiar with the mathematical formulation of general relativity: The description of the Einstein field equation given in this box corresponds to the mathematical relation $R_{tt} = 4\pi G(T_{tt} + T_{xx} + T_{yy} + T_{zz})$, where R_{tt} is the time–time component of the Ricci curvature tensor, G is Newton's gravitation constant, T_{tt} is the density of mass expressed in energy units (see Box 5.2), and $T_{xx} + T_{yy} + T_{zz}$ is the sum of the principal pressures along three orthogonal directions. See page 406 of MTW. This "time–time" component of the Einstein field equation, when imposed in *all* reference frames, guarantees that the other nine components of the field equation are satisfied.

119 [As I browse . . . (a browsing which, . . . into English!)J Einstein's personal papers and the rights to some of his published papers were tied up in a legal battle for several decades. The Russian edition of his collected works was produced and published at a time when the Soviet Union did not adhere to the International Copyright Convention. The far more complete English edition is now being published, very gradually; the first two volumes are ECP-1 and ECP-2.

CHAPTER 3

121 ["The essential result of this investigation . . . reality."] Einstein (1939).

122 [In 1783 John Michell . . . should look like.] Michell (1784). For discussions of this work see Gibbons (1979), Schaffer (1979), Israel (1987), and Eisenstaedt (1991).

123 [Thirteen years later, . . . subsequent editions of his book.] Laplace (1796, 1799). For discussions of Laplace's publications on dark stars, see Israel (1987) and Eisenstaedt (1991). Eisenstaedt discusses the attempts and failure to verify, observationally, Michell's prediction that light emitted by massive stars is affected by their gravitational pull, and the contribution that this failure might have had to Laplace's deletion of dark stars from the third edition of his book.

124 [Schwarzschild mailed to Einstein . . . curvature *inside* the star.] Schwarzschild(1916a,b).

131 [Jim Brault . . . Einstein's prediction.] Brault (1962). For a detailed discussion of tests of Einstein's general relativistic laws of gravity see Will (1986).

131–132 [However, few were . . . highly compact stars.] For a detailed discussion of the early history of people's reaction to the Schwarzschild geometry and research on it, see Eisenstaedt (1982). A broader-brushed history that covers the period from 1916 to 1974 will be found in Israel (1987).

135 [In 1939, Einstein published . . . cannot exist.] Einstein (1939).